

Optimizing the Transport of Organs for Transplantation

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ABSTRACT

As an organ becomes available for transplantation, a recipient must be selected. Usually, donor and recipient are geographically apart. Therefore, the transport of the organ must be planned and executed within the time window imposed by the maximum preservation time of the organ, which can impact recipient selection. The Cold Ischemia Time - CIT, that is the time elapsed between the surgical removal of the organ and its transplantation, must be the minimum possible to improve the transplantation success. In this sense, the air transport becomes the best option and, sometimes, it is the only way to deliver the organ before perishing. The planning of an organ transportation means choosing, among thousands of possible sequences of flights, the option that delivers the organ faster to its destination. This problem can be modeled as a resource constrained shortest path. Given the urgency and importance of this task, which is solved manually in Brazil, we present a labeling algorithm to find the optimal sequence of flights. Computational tests performed on 25 Brazilian real cases showed a reduction, on average, of 37,46% for the CITs and 44,17% for the transport times.

1. Introduction

Organ transplantation has saved and improved the quality of life of thousands of patients since the first successful kidney transplant in 1954. It is the best life-saving treatment for end-stage organ failure, and has been successfully performed in 111 countries (European Directorate for the Quality of Medicines & HealthCare, 2022). Organs can be donated from living or deceased persons with reported brain death, the latter being the major source of organs for transplantation. Even though one deceased donor can donate up to eight lifesaving organs, the demand is generally larger than the offer, and waiting lists continue to grow each year (Health Resources and Services Administration, 2022).

The transfer of human cells, tissues or organs from a donor to a recipient requires a balance between fairness and medicine to decide upon the effective recipient (Fuzzati, 2005). Certain medical factors are prioritized to determine if a person in the waiting list is a potential recipient: blood type, weight, height, age, etc. (United Network for Organ Sharing, 2019). Some of these characteristics can determine a definite incompatibility, e.g. blood type, while others, such as weight and height, may indicate how suitable the organ is for the recipient. Furthermore, patients with a higher urgency, with greater estimated chances of survival and benefit, normally appear on top when the

ranked list of candidates is generated (Fuzzati, 2005; United Network for Organ Sharing, 2019).

The idea of a static queue where a patient waits in line for an organ does not apply. Instead, each time an organ is available for transplantation a ranked list is generated based on the organ allocation policy. The differences in the level of policies, educational campaigns, governance, management systems within distinct regions and countries leads to a scenario where the rates of organ donation heavily varies around the world.

Besides organ allocation policies, the donation of an organ can represent a logistics challenge since organs lose viability (at an organ-specific rate) once they are removed from the donor's body (Alagoz et al., 2009). According to Health Resources and Services Administration (2021), each organ has its own cold ischemic time limit. While kidneys can survive outside the body for 24 to 36 h, the pancreas lasts on average 12 to 18 h, livers about 8 to 12 h, and hearts and lungs only 4 to 6 h.

In addition, the donor might be geographically far apart from the recipient, requiring a proper storage and a fast and reliable transport. In countries with continental dimensions such as Brazil, air transportation is often the only option capable of meeting the maximum preservation

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time constraint. Consequently, in 2001, the Brazilian Ministry of Health established a cooperation agreement with airports. The Brazilian Air Force and airlines were to transport tissues, organs and medical staff for transplantation purposes through military or commercial flights voluntarily and free of charge (Brasil, 2017, 2022).

To avoid confusion or misuse of some terms, some definitions follow. The Cold Ischemia Time (CIT) is the time interval between the blood supply cut off and its restoration in an organ or tissue (National Cancer Institute, 2022). As stated in Kayler et al. (2017b,a) and Ponticelli (2015), it varies with the time necessary to transport an organ for transplantation (Kayler et al., 2017b). The maximum preservation times are assumed to be organ dependent fixed values, for which they remain viable. These definitions will be used throughout the text.

We address the context for which organs are shipped from an origin airport and must arrive within the time window imposed by the maximum preservation time of the organ in a destination airport. However, the faster the organ arrives, the better transplantation outcomes tend to be due to lower cold ischemia times (Ponticelli, 2015). Furthermore, the transport should be performed with the least possible number of flights in order to minimize handling and exposure, maintaining the integrity of the organ and avoiding its contamination during transport. Li et al. (2022) support this need by discussing polymicrobial contamination of preservation fluids during organ procurement and transport as a significant problem, which poses risks to organ viability and recipient safety during kidney transplantation.

In Brazil, the National Transplantation Central (*Central Nacional de Transplantes - CNT*) is responsible for the procurement and distribution of organs and tissues for transplantation among the Brazilian states, managing and controlling receiver waiting lists at the state, regional and national levels (Brasil, 2022). Despite being a very complex and delicate task, the planning of organs' transport is performed manually by CNT technicians, potentially leading to an underutilization of the free transport agreement. Choosing an adequate trade-off between speed and reliability from a myriad of possible paths, each combining a subset of hundreds of possible flights, is a rather daunting – not to mention unfair – task.

Software support systems might help speed up and simplify some of the operations of the organ procurement phase, guaranteeing a better use of resources and increasing the chances of success (Fuzzati, 2005). In order to automate the CNT technicians' work, which must be completed in minutes, and mathematically determine the shortest path between origin and destination airports, Carrara et al. (2018) proposed a Mixed Integer Linear Programming (MILP) model to optimally plan the transport of organs for transplantation. These authors showed that the mathematical model could reduce the transport times and result in a more fair choice of the recipient, when compared to the real decisions previously taken.

In addition to its practical relevance, their work is also to be praised due to its novelty. This is because the scientific literature on operations research applied to the transplantation of organs has mainly focused on designing location-allocation of health care facilities or on optimizing organ transplant supply chains (Mantecchini et al., 2016). However, for some instances, Carrara et al. (2018) reports that a commercial solver could not prove optimality or even find a feasible solution in ten minutes of execution, the amount of time available to plan the organ transportation.

The MILP model proposed by Carrara et al. (2018) falls into the shortest path problem category, since the nodes and arcs can be used as abstractions for airports and flights. However, the constraint imposed by the maximum preservation times should not be ignored, leading to a resource constrained shortest path problem (RCSP), which according with Irnich and Desaulniers (2005) can be solved using pseudo-polynomial algorithms.

In this work we model the transportation of organs for transplantation problem as a RCSP, and we present an implementation of a dynamic programming labeling algorithm to optimally, and efficiently,

solve it. The dynamic programming solution approach can be simply understood as breaking a problem into smaller parts, which have the property of being themselves optimal, and solving this parts recursively (Bellman, 1954). Labels are used to store information, such as time and distance, from a path arriving at a given node (Desrochers and Soumis, 1988). Additionally, since a previous attempt in the literature failed at optimally solving larger instances through the use of a commercial solver, bridging this gap justifies this research.

The remainder of this work is organized as follows. Section 2 presents a brief review on applications of operations research in the organs transplantation context. Section 3 describes the MILP model proposed in Carrara et al. (2018). Section 4 shows the solution methodology of the problem variant and presents the pseudocode of the algorithm implemented. Section 5 address the performance of the algorithm and the results obtained. At last, Section 6 draws conclusions.

2. Literature review

The research of operations has been widely applied in the healthcare domain. The applications range from optimal location of hospitals and emergency vehicles, patient and medical staff scheduling, to disease diagnosis, as seen in Rais and Viana (2010). In this section, a literature review on optimization applied to organ transplantation is provided, considering operations research approaches and, in particular, the shortest path problem.

2.1. Operations research approaches for organ transplantation

In the organ transplantation context, issues such as system's efficiency and allocation's fairness naturally arise. We observed that most authors in the literature attempt to address these questions through location-allocation models and discussions regarding organ allocation policies.

A model aimed at optimally organizing an organ transplant system is introduced in Bruni et al. (2006). The authors analyze the Italian case, where different regions of the country have different waiting times. The location-allocation model proposes a reorganization of the transplant system and an augment on regional equity. This happens through the selection of a set of locations that will result in the shortest maximum waiting list for all regions. The model was experimentally validated in the Italian transplantation system, showing a potential to better spatially distribute transplant centers.

In Beliën et al. (2013), a MILP facility location model is proposed with the aim of minimizing the waiting time from the moment an organ becomes available until implantation, by looking for the optimal set of transplant centers to open for each organ. This model also considers the waiting time until the organ removal surgery, which is less important, and therefore, should have reduced weight in the objective function summation. The model was applied to different scenarios based on Belgian real data from 2004–2009. The authors conclude that if the objective function aims at minimizing only the CIT, few transplant centers are opened, leading to a centralization scenario. However, if the total time is minimized, there is a trend to open many transplant centers, leading to a decentralized scenario. They also report that all instances were solved to optimality with very small computational times.

A location-allocation MILP model to minimize the total costs and elevate the efficiency of an organ transplant network in Tehran is developed in Zahiri et al. (2014). The deterministic model proposed is compared to a robust approach, which considers uncertainty. The models are tested with the Iranian organ transplant supply chain real-world data based on different scenarios. The authors suggest converting the model into a bi-objective MILP to minimize the total traveled time as the second objective, extending the model to greater scales and using metaheuristic algorithms to solve it.

The effects of cluster structures on potential compatible donor-recipient matches within the cold ischemia time bounds are investigated in Savaşer et al. (2019). Based on a case study performed in Turkey, the research proposes mathematical models to maximize intra-regional transplant flow by exploring the optimal clusters for different transportation options as a strategic level decision. The results highlight that, by decreasing the transportation time of an organ, the time available for surgery increases as well as its performance. A possibilistic programming model was developed by Kargar et al. (2020) for a liver transportation and allocation problem, considering medical uncertainty and trade-off between quality metrics, namely efficiency and equity. For this purpose, a multi-objective optimization model with four objectives is presented: i. to increase the survival rate of eligible patient-organ pairs; ii. to meet the urgent needs of a patient; iii. to minimize the total transportation costs occurring in the transplant network; and iv. to increase the possibility of patient access to the nearest transplant centers. Rouhani and Amin (2022) deals with the efficient and timely management of organ donations from the perspective of supply chain network design. A novel bi-objective hierarchical location-allocation is presented, considering the reduction of time and total costs and the maximization of geographical parity in the network under supply and demand uncertainty.

Cacchiani et al. (2018) present a model to optimize the distribution of aircraft in a set of hubs over Italy, which falls in the uncapacitated facility location category. Instances are based on the Italian database from June 2015 to May 2016. Two scenarios were modeled: two hubs and three hubs. Six aircraft were necessary to cover transport requests in both scenarios. The authors concluded that a larger number of hubs would allow a reduction in the total distance flown, and consequently less fuel consumption and polluting emission.

An analysis of the Italian organ transport logistics chain is presented by Mantecchini et al. (2016). All transport activities over 44 Italian transplant centers and the related airport network were monitored in real-time, investigating parameters such as origin and destination of the organ, transport type, times, etc. The data was collected between June and July 2015 and corresponds to 128 organ transport events. A further study containing Italian data between June 2015 and July 2016 is presented in Mantecchini et al. (2017). In this period a total of 617 organs were transported by air and in 417 cases, the organs were accompanied by medical staff.

Different of the studies aforementioned, this work extends the traditional location-allocation approaches by adapting the shortest path problem with resource constraints to optimize the transport of organs for transplantation. The literature for this problem is highlighted in the following subsection.

2.2. The resource constrained shortest path problem

According to Gallo and Pallottino (1986), all solution algorithms for the shortest path problem are derived from a single procedure, which corresponds to a dynamic programming algorithm. The algorithm is capable of finding the optimal solution for the shortest path problem through a recursion, building paths from the origin node s to the destination node t (Desrochers and Soumis, 1988; Ioachim et al., 1988; Wolsey, 1998). The effectiveness of dynamic programming algorithms to solve the shortest path problem relies on the “principle of optimality”. This corresponds to the property of having an optimal substructure, where pieces of an optimal solution are themselves optimal (Wolsey, 1998). Core references for the shortest path problem in the literature are Prim (1957), Kruskal (1956), Bellman (1958), Ford and Fulkerson (2015) and Dijkstra (1959).

In a classical shortest path problem formulation, the cost of a solution is equal to the sum of the costs of the edges used to traverse the optimal path from the source to the sink node. These costs can be seen as lengths, or in a more general sense, they can be interpreted as the consumption of a resource. A resource corresponds to a quantity

that varies along the path, such as distance, time, and load (Irnich and Desaulniers, 2005).

The shortest path problem has variants and, in these cases, a set of constrained resources that are known as the *shortest path problem with resource constraints* - SPPRC needs to be dealt with. As in the unconstrained version, paths are built in a stepwise approach but, this time a multi-dimensional resources vector is accumulated and constrained at each node, introducing the concept of feasibility. It is necessary to consider all incomparable paths arriving at a given node, since a path “a” can be better than a path “b” regarding the consumption of a resource “r”, and the same path “a” can be worse than path “b” regarding the consumption of a different resource “s”. This observation provides an initial insight into the complexity to solve SPPRC complexity (Irnich and Desaulniers, 2005).

An early attempt to solve a side constrained shortest path problem was presented in Jokschi (1966), in which arcs were characterized by cost and a non-negative second variable. In this case, the optimal path being the most economical that satisfies a constraint requiring that the sum of these arc’s second variables must be greater than or equal to a defined parameter. In a similar sense, Handler and Zang (1980) and Aneja et al. (1983) present a shortest path formulation with an additional knapsack constraint and address the problem with Lagrangean relaxation and implicit enumeration algorithms, respectively. In Handler and Zang (1980), the authors suggest that this additional constraint can be interpreted as a total time constraint in a transport network. This points out in the direction of what perhaps is the most valuable resource, and therefore, the one which motivates most researches and applications: time.

The *shortest path problem with time windows* - SPPTW was first studied in Desrosiers et al. (1983). It is a time-constrained variant of the shortest path problem with time window constraints at each node. The time windows are an efficient way to model allowable delivery times of customers in many routing and scheduling problems (Desrochers, 1986). The problem was generalized and addressed with an algorithm in Desrochers and Soumis (1988) for the case with several constrained resources (SPPRC).

The resource-constrained shortest path problems are very common as subproblems in several column generation and branch-and-price schemes. They are used to solve routing and scheduling problems, having contributed to the success of these methods. In these cases, they work as a flexible tool to model cost structures and feasibility rules of routes and schedules, since there are efficient algorithms for its most important variants (Irnich and Desaulniers, 2005). In respect to efficiency, the permanent labeling algorithm, proposed by Desrochers and Soumis (1988), is said to run in pseudo-polynomial time. This algorithm solves instances with up to 2500 nodes and 250.000 arcs. This kind of mono-directional algorithm starts with an empty and trivial path at the source node and calculates labels iteratively as paths are built.

More recently, Righini and Salani (2006) presented a bounded bi-directional dynamic programming algorithm for the *elementary shortest path problem with resource constraints*. In this case, in order to minimize the growth of the number of labels along the path construction, a forward and a backward path are built from the source and sink node, respectively. After that, these two paths are joined together. Their bi-directional implementation has been shown to outperform the mono-directional one. In the same effort to minimize labels growth, Tilk et al. (2017) experimentally observed that the forward and backward label extensions are unbalanced. Therefore, the authors proposed a dynamic half-way point based on the current state of the solved forward and backward paths. The dynamic half-way point implementation has shown to speed up the computational time in up to 41% when compared with the previous static bi-directional implementation.

Other studies corroborate the efficiency of dynamic programming algorithms for the shortest path problem with limited resources. In

Ahmadi et al. (2021), the authors introduced heuristics that significantly improve the initialization phase performance, considering the bidirectional A* algorithm. Their efforts focus mainly on improving the search for lower bounds. Cabrera et al. (2020) proposed an exact bidirectional pulse algorithm for the constrained shortest path problem. This method is based on an adjustable depth-first (or breadth-first) search executed in parallel both from the starting and ending node, as well as a set of acceleration strategies. The authors incorporated core pruning strategies that allow to prune partial paths at early stages of the search, showing good results regarding optimality and computational time.

Ahmadi et al. (2024) presented two new solution approaches for the weight constrained shortest path problem. The first one it is an unidirectional algorithm derived from techniques used in bi-objective search algorithms, while the second is an extended version of the latest-generation bidirectional constrained search algorithm, enhanced with parallelism. The research results reveal significant improvements in both runtime and memory requirement to previous algorithms, such as the bidirectional pulse algorithm. The authors conducted an extensive analysis and compared the performance of the algorithms with recent approaches, demonstrating the advantages of the proposed methods.

Shortest path algorithms applied to healthcare are not a novel approach. A shortest path analysis of the spatial accessibility of healthcare services in the Sichuan Province can be found in Pan et al. (2015). This method, which was implemented in a GIS-based environment, represents a more sophisticated analysis in comparison with the current regional availability approach used by policy makers, which solely calculates the ratio between population and healthcare services within administrative boundaries. According to the authors, this approach could provide useful information for healthcare planning and public health policies, as it identifies that the accessibility is highly uneven throughout the province. In the same sense and also in China, Zhu et al. (2019) presents an evaluation of the spatial accessibility to beds, doctors and nurses in Shenzhen. The analysis was performed, among others, with the shortest path method, which was used to examine the geographical potential of hospital utilization.

A Bellman–Ford implementation to solve a sequence of shortest path problems that arise in the pricing problem of a column generation scheme can be found in Ares et al. (2016). The Column Generation algorithm aims to solve a set-partitioning formulation to locate a given number of roadside clinics in Africa. The mathematical model aims to provide equal access for truck drivers along different truck routes in Sub-Saharan Africa, since they should be sufficiently close to these facilities at every moment during their trips in order to their treatments to be effective. Computational tests were performed in 59 randomly generated instances, and the authors report near-optimal solutions within an acceptable amount of time for large sized instances, outperforming a previously direct approach. In addition, they report that considerable gains in terms of equity can be achieved.

An stochastic shortest path model to find an optimal sequence of tests to confirm or discard a disease, regarding an optimal testing policy, can be found in Arruda et al. (2018). The model proposed in this work takes Bayesian statistics to, after one test, sequentially derive the posterior probability of a disease. The authors report that the model is related to sequential hypothesis testing, but with fundamental differences, such as a limited number of tests, each can be applied just once, and an individual cost for each test, thus not imposing any constraint in the cost function. The model is applied to compare tests for Coronary artery disease. Tests for an optimal costs policy and a optimal diagnosis policy are performed, showing a small difference in the probability of a correct diagnosis, but larger differences regarding to costs. Although there is no official guideline in Brazil, the authors report that the consensual strategy among physicians is to prioritize costly tests, in a similar diagnosis policy fashion. The authors affirm that the model can be applied to evaluate new technologies for disease detection.

Although organ transportation is tangent to these models, an approach which solely focuses on the transportation of organs through regions can be found in Carrara et al. (2018). The authors propose a model to optimize the transport of organs by air through the utilization of commercial flights in Brazil. The model was solved with CPLEX in instances based on data regarding 25 transplanted organs, collected from the CNT. The execution times vary from 5 s to 10 min (the maximum execution time allowed). The authors suggest that a shortest path with resource constraints algorithm could reduce the execution time and ensure optimality for all instances.

In this work we extend the study of Carrara et al. (2018) by modeling the transportation of organs for transplantation problem as a resource constrained shortest path problem, and presenting an implementation of a dynamic programming labeling algorithm to optimally, and efficiently, solve it. More specifically, we show that our algorithm can compute the shortest path between all airports providing the optimal path (if one exists) with low execution times. The relevance of this research also lies in its potential to avoid additional costs, such as those associated with the acquisition of commercial solver licenses by the CNT.

According to the literature, there are many efficient algorithms to solve the restricted shortest path problem. However, for this research, we chose a dynamic programming labeling algorithm due to its ability to handle multiple resources simultaneously. Thus, in order to obtain optimal answers as fast as possible, our algorithm takes advantage of the constrained nature of the problem and optimizes solutions regarding two key resources: time and flights. Considering a multidimensional resource vector, and ensuring feasibility at each node, our approach extends traditional shortest path algorithms to handle complex resource constraints effectively. We propose an efficient and fast algorithm for the complex organ transport logistics problem.

3. Mathematical formulation

This section introduces the Mixed-Integer Linear Programming model proposed in Carrara et al. (2018), designed to optimally solve the transport of organs for transplantation purposes. In addition to some additional constraints, this mathematical formulation is similar to the shortest path problem, one of the most simple and applicable problems in combinatorial optimization (Wolsey and Nemhauser, 1999).

Let $G = (\mathcal{V}, \mathcal{A})$ be a directed graph, in which $\mathcal{V}$ and $\mathcal{A}$ correspond to the sets of nodes (airports) and arcs (flights), respectively. The airport of origin (source node) is denoted by s and the destination airport (sink node) is denoted by t , while the set of candidate airports to intermediate stops, or transshipment nodes, is denoted by $\mathcal{V}_r$. Thus, $\mathcal{V}$ can be rewritten as $\mathcal{V} = \mathcal{V}_r \cup \{s, t\}$. Let $\mathcal{A}_{ij}$ be the set of all arcs available from a node $i \in \mathcal{V}$ to a node $j \in \mathcal{V} : i \neq j$. Thus, $\mathcal{A}$ can be written as $\mathcal{A} = \bigcup_{(i,j)} \mathcal{A}_{ij}, \forall i, j \in \mathcal{V} : i \neq j$.

For each arc $a \in \mathcal{A}_{ij}$, which corresponds to a flight from i to j , there is an associated departure time h_{ij}^a and a flight duration d_{ij}^a . According to the organ to be transplanted, there is a period of time D_{max} to transport the organ from the origin airport s to the destination airport t , which is related to the maximum preservation time of the organ. The organ is ready to be transported from time $H_{available}$ and must arrive in the airport t in a moment T_t not superior to the maximum possible time $H_{t,max}$.

When a stop in one or more airports from the set $\mathcal{V}_r$ is made, it must be guaranteed that there is enough time to handle the organ in an appropriate manner when an exchange of aircraft is necessary. The time required for these operations is represented by parameter τ_{ij} . Carrara et al. (2018) uses 30 min for all pairs of airports.

Even with the accounting of this handling time by the formulation, ideally, stopovers should be avoided to reduce the manipulations of the organ and also the possibilities of unforeseen events. In order to choose a solution with a good balance between the earliest arrival time at the

destination airport and the fewest number of flights along the path from s to t , a penalty P , adjustable by the modeler, is added.

In respect to the decision variables, let $T_i \geq 0$ represent the time when the organ leaves node $i \in \mathcal{V} \setminus \{t\}$. When the airport into consideration is the origin s , it must be guaranteed that the flight chosen leaves the airport after the organ is available for transport, $T_s \geq H_{available}$. Thus, let x_{ij}^a be a binary decision variable that assumes 1 if the arc (flight) $a \in \mathcal{A}_{ij}$ is chosen to make the trip between the nodes $i \in \mathcal{V}$ and $j \in \mathcal{V}$, and 0 otherwise.

Taking into account the definitions above, the mathematical model is presented below.

$$\text{Minimize } z = T_t + P \sum_{i \in \mathcal{V} \setminus \{t\}} \sum_{j \in \mathcal{V} \setminus \{s\} : i \neq j} \sum_{a \in \mathcal{A}_{ij}} x_{ij}^a \quad (1)$$

Subject to:

$$\sum_{j \in \mathcal{V} \setminus \{s\}} \sum_{a \in \mathcal{A}_{sj}} x_{sj}^a = 1 \quad (2)$$

$$\sum_{i \in \mathcal{V} \setminus \{t\}} \sum_{a \in \mathcal{A}_{it}} x_{it}^a = 1 \quad (3)$$

$$\sum_{i \in \mathcal{V} \setminus \{j,t\}} \sum_{a \in \mathcal{A}_{ij}} x_{ij}^a = \sum_{i \in \mathcal{V} \setminus \{j,s\}} \sum_{a \in \mathcal{A}_{ji}} x_{ji}^a \quad \forall j \in \mathcal{V}_r \quad (4)$$

$$T_j \geq T_i + d_{ij}^a + \tau_{ij} - M_{ij}(1 - x_{ij}^a) \quad \forall i \in \mathcal{V} \setminus \{t\}, j \in \mathcal{V} \setminus \{s\}, a \in \mathcal{A}_{ij} \quad (5)$$

$$T_i = h_{ij}^a x_{ij}^a \quad \forall i \in \mathcal{V} \setminus \{t\}, j \in \mathcal{V} \setminus \{i\}, a \in \mathcal{A}_{ij} \quad (6)$$

$$T_s \geq H_{available} \quad (7)$$

$$T_t \leq H_{t,max} \quad (8)$$

$$x_{ij}^a \in \{0, 1\} \quad \forall i \in \mathcal{V} \setminus \{t\}, j \in \mathcal{V} \setminus \{i\}, a \in \mathcal{A}_{ij} \quad (9)$$

$$T_i \geq 0 \quad \forall i \in \mathcal{V} \setminus \{t\} \quad (10)$$

The objective function minimizes the sum in (1), which corresponds to the arrival time at the destination airport T_t plus the number of flights taken multiplied by a penalty P . Constraints (2) and (3) are responsible for the outflow and inflow at the source and sink nodes, respectively, guaranteeing that a flight leaving s and a flight arriving at t are going to be selected. The set of Constraints (4), on the other hand, ensures the conservation of flow at the transshipment nodes. The set of Constraints (5) is responsible for the right accounting of time, ensuring that the organ is available for transport from a destination j at least τ_{ij} minutes after its arrival, destined for the handling of the organ. These constraints are in a linearized form, considering the Miller–Tucker–Zemlin subtour elimination constraints (Miller et al., 1960), in which M_{ij} is a sufficiently large constant. Constraints (6) set the departure time at a given node equal to the departure time of the flight, i.e. arc, chosen to leave the node. Constraint (7) ensures that the organ leaves the origin airport only after the moment it is available for transport, while Constraint (8) ensures that the organ must arrive at the destination t before the maximum arrival time. This pair of constraints over the resource time leads the problem to a different shortest path variation, namely the resource constrained one. At last, Constraints (9) and (10) are associated with the decision variables domains.

This model yields the shortest path between two airports, also considering the number of flights necessary to traverse this path. The answer consists of a sequence of flights within the time window imposed by the maximum preservation time of the organ transported. However, as reported in Carrara et al. (2018), the solution of the model becomes complex if the number of flights and nodes increase.

A dynamic programming algorithm is presented in the next section. This algorithm was developed to mitigate the effects caused by the growth of the number of decision variables. This approach was considered since the processing time is usually limited to ten minutes.

4. Solution methodology

This section presents the labeling algorithm to solve the transport of organs for transplantation problem. The differences on unconstrained and resource constrained versions are explained briefly, and the most recent attempts to solve its constrained version efficiently are presented. Then we explain how the practical problem here addressed fits into the resource constrained shortest path problem. At last, necessary mathematical definitions are made and the algorithm is formally presented.

The algorithm uses the same directed graph $G = (\mathcal{V}, \mathcal{A})$ introduced in Section 3, in which $\mathcal{V}$ corresponds to the nodes (airports) and $\mathcal{A}$ corresponds to arcs (flights). There are different flights (multiple arcs) between two airports (nodes) over the day, thus it is not possible to represent a path by a sequence of nodes. A path $P = (a_0, \dots, a_p)$ with length p is represented by a sequence of arcs (flights) where the arrival airport (head node) of $a_i \in \mathcal{A}$ has to be equal to the departure airport (tail node) of $a_{i+1} \in \mathcal{A}$ for all $i = 0, \dots, (p-1)$. There are no path-structural constraints, see Irnich and Desaulniers (2005) for more details, thus all paths are structurally feasible.

The maximum preservation time of the organ to be transported imposes a viability time window. Time is, therefore, the most important resource for this application, and all nodes are constrained by the time window $[H_{available}, H_{t,max}]$. Another resource taken into consideration is the number of flights used from the origin airport s to the destination t . Despite being an unconstrained resource, a good trade-off between the number of flights and the arrival time at t can be interpreted as an attempt to avoid unforeseen events and minimize the necessity of handling the organ, which is clinically not recommended. This trade-off is modeled through the use of a penalty P at the pricing of the solution, following the objective function (1) from Section 3.

The SPPRC variant for the transport of organs problem deals with two resources: *time* and *flights*. Thus, a label (time, flights) is associated for each feasible path P_{sj} from the origin s to a node j , storing the arrival time at the node j and the number of flights taken up to j . A label representing a path P_{sj} from the origin s to a node j will be denoted by (T_j^k, F_j^k) , in which k corresponds to the k th path from s to j . The number of labels k stored at each node is equal to the number of efficient labels arriving at j .

For two paths P_{sj}^1 and P_{sj}^2 from s to j with two respectively associated labels (T_j^1, F_j^1) and (T_j^2, F_j^2) , P_{sj}^1 is said to dominate P_{sj}^2 if and only if $T_j^1 \leq T_j^2$ and $F_j^1 \leq F_j^2$. For a given node j , a label (T_j^k, F_j^k) is said to be efficient if no other label at j dominates it. Similarly, a path P_{sj}^1 is said to be efficient if its associated label (T_j^1, F_j^1) is efficient.

Non-efficient labels can be discarded in a label treatment step. However, even treating labels and discarding some of them, their number can grow rapidly and increase the processing time of the algorithm. This happens because labels represent paths that have to be extended until the destination t is reached. The way how paths are extended and resource consumption is accumulated throughout the path construction depends on the resource extension functions defined.

A *resource extension function* - REF $f_{ij}^r : \mathbb{R} \rightarrow \mathbb{R}$, defined over a resource r , depends on the consumption of the resource r accumulated along the path from the origin s until the node j and normally extends the consumption of this resource with the amount used at arc (i, j) . For each resource taken into account in our implementation, i.e. *time* (T) and *flights* (F), a REF is defined as follows:

$$f_a(T_j^k) = h_{ij}^a + d_{ij}^a \quad \forall i \in \mathcal{V} \setminus \{t\}, j \in \mathcal{V} \setminus \{s\}, a \in \mathcal{A}_{ij}, k \in \mathbb{N} \quad (11)$$

$$f_a(F_j^k) = F_i^k + 1 \quad \forall i \in \mathcal{V} \setminus \{t\}, j \in \mathcal{V} \setminus \{s\}, a \in \mathcal{A}_{ij}, k \in \mathbb{N} \quad (12)$$

The time resource extension function (11) calculates the arrival time at j by summing up the departure time from a previous node i , h_{ij}^a , with the duration of the flight from i to j , d_{ij}^a . The usage of the resource time at a given label k of j , T_j^k , which corresponds to the arrival time at j , must lie within the time window $[H_{available}, H_{t,max}]$. Otherwise, the path

extension from a previous node i towards j is not feasible and will not be performed. The flights resource extension function (12) calculates the number of flights used at a given label k of j by adding 1 to the total number of flights used at the label k from the previous node i .

The time window $[H_{available}, H_{t\ max}]$ defined over all nodes is enough to satisfy the sets of constraints (7) and (8) from the mathematical formulation presented at Section 3. However, it is not enough to satisfy the set of constraints (5), which requires that the organ is available for transport from a transshipment node $i \in \mathcal{V}_T$ to node $j \in \mathcal{V}_R$, only τ_{ij} minutes after its arrival. In order to expand paths and keep them feasible in respect to these three sets of constraints, flights with departing time that do not satisfy this constraint will be discarded through preprocessing.

Besides embedding constraints (5), (7), and (8), in the Preprocessing Step we are also able to choose a single feasible and dominant arc (when there is at least one feasible arc) among all arcs available to expand the path P_{si}^k until P_{sj}^k . We can observe easily that the number of flights F_j^k will be the same for a path extension from i towards j , arising from the path $P_{si}^k = (T_i^k, F_i^k)$, since it depends only on F_i^k . A feasible flight, i.e., one that satisfies the sets of constraints, can be observed. That results in the earliest arrival time T_j^k that would generate a dominant label $P_{sj}^k = (T_j^k, F_j^k)$, when compared with the other feasible arcs. Thus, being not necessary to consider more than just one arc for each path extension between two nodes.

A path extension in all directions, i.e., towards all nodes $v \in \mathcal{V}$, can be understood as processing a path, or treating a label. Let $\mathcal{U}$ be the set of *untreated labels* and $\mathcal{P}$ be the set of *treated labels*. The main aspect of labeling algorithms is an efficient manipulation of these two sets. Since the number of labels can grow rapidly, it is also important to apply dominance rules and discard non-efficient labels. As labels are extended and discarded throughout the execution of the algorithm, the sets $\mathcal{U}$ and $\mathcal{P}$ change dynamically. Briefly explained, the algorithm starts with the trivial path P_s^0 in the unprocessed set $\mathcal{U}$, and the set $\mathcal{P}$ empty, and terminates when there are no more labels to treat, i.e., the set $\mathcal{U} = \emptyset$.

The Algorithm 1 is heavily based on Irnich and Desaulniers (2005), with adjustments to the organs transport variant, previously described. The code was fully implemented in C programming language and the data structure used to represent the paths was a multi-dimensional array of the nodes. To ensure optimality, it is necessary to store an unknown Ω number of efficient labels at each node, which can be achieved through a linked list implementation or a sufficiently large number of empty positions.

The dominance step can be applied at every iteration or be delayed to a point when there is a chance to remove several non-efficient labels before they are processed in the path extension step. In a first implementation, we applied the dominance step at every iteration, but the results were not good. So, to keep a good performance, the strategy adopted was to embed the dominance step into the Path Extension Step (see Algorithm 1). This was achieved by comparing the candidate label with all labels already stored in the node the path is moving towards.

As paths are extended in the Path Extension Step, if the candidate label dominates a previously stored label, it replaces the latter, which is discarded. Otherwise, if the candidate label is dominated by a previously stored label, the former is discarded. Finally, if the candidate label neither dominates nor is dominated by a previously stored label, it is stored at the corresponding node.

Since there is no trigger point in the algorithm to terminate the execution when one or more feasible paths arriving at the destination node t are found, the complete execution of the algorithm calculates the shortest paths between all nodes $v \in \mathcal{V}$. In the Pricing Step, the algorithm prices all paths arriving at the destination node t when all paths are evaluated. However, the algorithm could return the shortest path from s to any other node without having to be recalculated.

This is specially useful when a list of destinations sorted in order of priority is provided. And, in the absence of a feasible path towards the first node of the list, the algorithm jumps to the next candidate

destination until a feasible path is found or until there is no more candidate destination. This could happen in a very constrained scenario, for example, when an organ with a short maximum preservation time imposes a narrow time window above all nodes and the priority receptor is geographically very far from the donor.

The pseudo-code presented in Algorithm 1 gives a better idea on the functioning of the algorithm developed in this work to find the best flights for the transport of organs.

Algorithm 1: SPPRC Labeling Algorithm

```

1 /* Initialization Step */
2 SET  $\mathcal{U} = \{P_s^0\}$  and  $\mathcal{P} = \emptyset$ 
3 while  $\mathcal{U} \neq \emptyset$  do
4   CHOOSE the least cost path  $P \in \mathcal{U}$  and REMOVE  $P$  from  $\mathcal{U}$ 
5   forall  $v \in \mathcal{V}$  do
6     /* Preprocessing Step */
7     FIND a feasible and dominant arc  $a^* \in A_{ij}$  towards  $v$ 
        from the head node of  $P$ , when there is at least one
8     /* Path Extension Step */
9     if  $\exists a^*$  then
10      EXTEND  $P$  towards  $v$ 
11      if the resulting path  $P_{sv}^k$  is dominated by a
        previously stored label  $P_{sv}^i : i \in \{0, \dots, \Omega\}$  then
12        DISCARD  $P_{sv}^k$ 
13      end
14      if the resulting path  $P_{sv}^k$  dominates a previously
        stored label  $P_{sv}^i, i \in \{0, \dots, \Omega\}$  then
15        DISCARD  $P_{sv}^i$ , STORE the resulting path  $P_{sv}^k$  and
        ADD  $P_{sv}^k$  to the set  $\mathcal{U}$ 
16      end
17      if the resulting path  $P_{sv}^k$  neither dominates nor is
        dominated by a previously stored label  $P_{sv}^i$  then
18        STORE the resulting path  $P_{sv}^k$  and ADD  $P_{sv}^k$  to
        the set  $\mathcal{U}$ 
19      end
20    end
21  end
22  ADD  $P$  to the set  $\mathcal{P}$ 
23 end
24 /* Pricing Step */
25 PRICE all  $\Omega$  paths  $P_{st}^k \subseteq \mathcal{P}$  arriving at the destination node  $t$ 
    and RETURN the optimal, i.e. least cost path  $P_{st}^*$ , when there
    is at least one

```

5. Computational results

The algorithm presented in the previous section was tested on real instances introduced in Carrara et al. (2018). The results obtained are compared with those in Carrara et al. (2018) for the same set of instances. Furthermore, a comparison between the routes taken on these real cases and the ones obtained with the aid of the dynamic programming algorithm is presented. Moreover, the effect of the penalty in the solutions is shown.

5.1. Case study

The algorithm was written in C, using the gcc 5.4.0 compiler with -O3 option. We used a computer with an AMD Athlon™ 64 X2 6000+ 3.0 GHz dual core processor, 8.0 Gb of RAM, and Linux Ubuntu 16.04.11 LTS operating system. The results from Carrara et al. (2018) were obtained in up to 10 min of execution of the formulation (1)–(10) by the solver ILOG CPLEX 12.5 using a computer with an Intel® Celeron® M 540 1.81 GHz processor, 2.0 Gb of RAM, and Windows 7 operating system. The MFlops relation between the two computers is

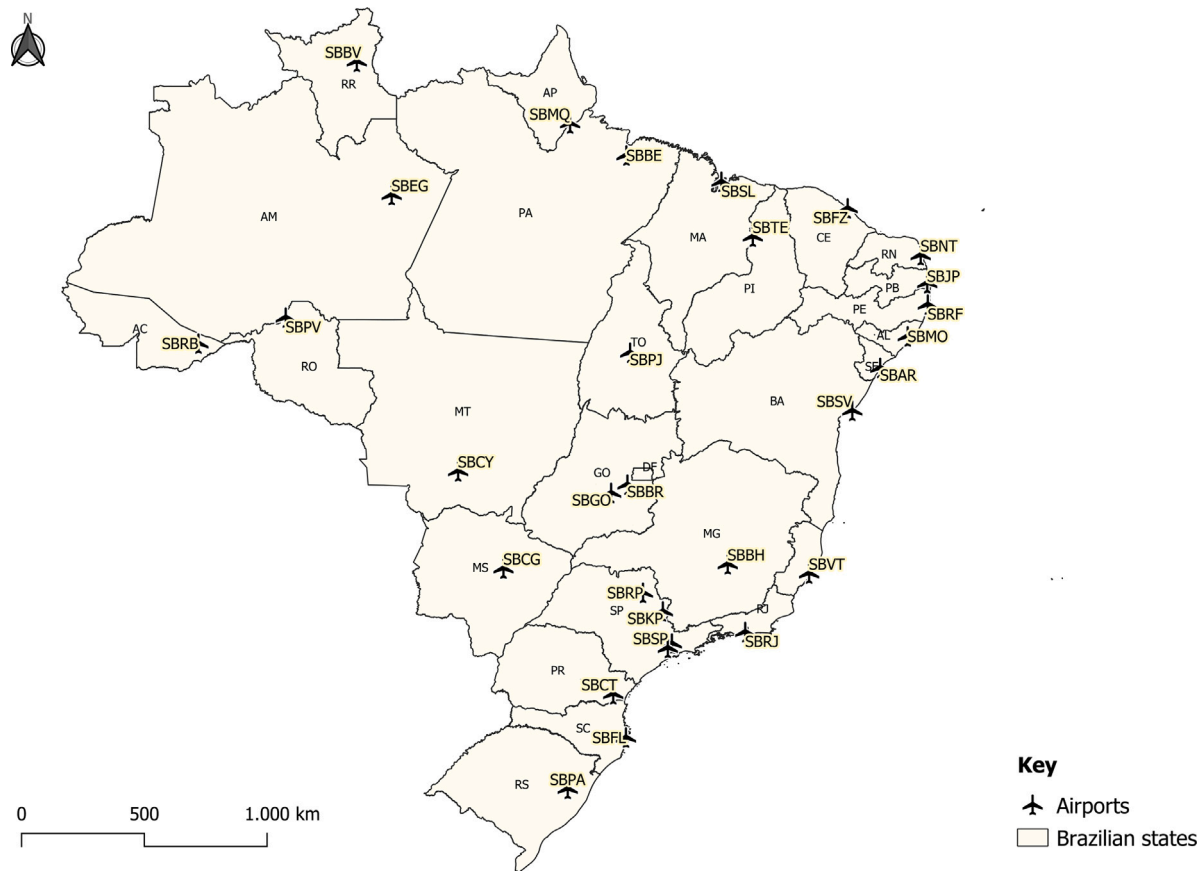


Fig. 1. The airport network.

approximately equal to 0.3 ($1.81/2 \times 3.0$). Thus, in order to fairly compare the results, the computational times of Carrara et al. (2018) were multiplied by this factor. All the computational times are expressed in seconds.

The instances are based on real data collected in Carrara et al. (2018) with the CNT. Each instance corresponds to a real case in which an organ was available for transplantation and a list of possible destinations, in order of priority, was provided. The airport network is composed by 32 Brazilian airports, one airport for each state capital plus some airports considered relevant, for example, an airport base of a Brazilian airline. The flight network is composed of all commercial flights operated by Brazilian airlines between these 32 airports. Santos Dumont and Galeão airports were treated as a single node in Rio de Janeiro (SBRJ), while Pampulha and Confins airports were considered as a single node in Belo Horizonte (SBBH). Fig. 1 illustrates the airport network.

The instances names are composed by a C letter followed by two numbers representing the number of the case (e.g. C01). For each case, instances were created from the priority destination until the real destination chosen. The [Table 1](#), which is adapted from [Carrara et al. \(2018\)](#), illustrates these cases and shows in red the destinations chosen by CNT.

The destination choosing task was performed manually by CNT technicians as they search for flights on the major Brazilian airlines websites, which can easily incur in a not fair choice in respect to priority and a suboptimal route. In fact, one can observe in [Table 1](#) that in twelve cases the first priority receiver was not contemplated. This reality, however, can be changed by the algorithm results.

5.2. Results obtained with the labeling algorithm

It is necessary to state that the answers presented in the following tables were obtained with the penalty parameter P equal to 30 min before presenting the results.

In addition, the times involved in the transplantation process have to be defined. Let D_{cg} be an organ dependant time required to perform the removal surgery. Let D_{hs} be the time necessary to transport the organ from the donor's hospital to the origin airport s , and D_{th} be the time necessary to transport the organ from the destination airport t to the recipient's hospital. Let D_{st} be the time elapsed since the moment an organ is available for transport at the origin airport s ($H_{available}$) until its arrival in the destination airport t (T). Let D_{max} be the maximum time available to perform the transport from airport s to airport t . Therefore, the following inequality holds: $D_{st} \leq D_{max}$. Finally, let the maximum cold ischemia time - CIT_{max} be a variable which corresponds to the organ maximum preservation time. The values assumed for this parameter are presented in [Table 2](#), as in [Carrara et al. \(2018\)](#).

The [Table 3](#) shows the comparison between the paths built by CNT and those provided by the labeling algorithm for the same destination, with differences highlighted in bold and red. The column Difference (Diff.) shows the difference in the transport times (D_{st}) between routes built manually and the ones provided by the algorithm. In cases where the transport was made by a charter aircraft or by an aircraft from the Brazilian Air Force (FAB), it is not possible to calculate this difference. However, except for case C08, it can be observed that it would be possible to perform this transport with the aid of commercial flights. In addition, the CIT reduction column shows the percentage gain in the cold ischemia time. In cases where it is possible to compare paths and times, the algorithm has been shown to reduce the CIT by 37,46% on average. And, when only the time to transport the organ from the

Table 1
Cases and destination chosen by CNT among a ranked receivers list.

Case	Organ	Origin	Priority order of destination								
			1st	2nd	3rd	4th	5th	6th	7th	8th	9th
C01	Kidney	SBSV	SBRF	SBFZ	SBRJ	SBVT	SBBH				
C02	Kidneys	SBRJ	SBVT	SBBH	SBRF	SBCT	SBPA				
C03	Kidneys	SBFZ	SBRF	SBRJ	SBVT	SBBR					
C04	Kidney	SBSV	SBRF	SBFZ	SBPA	SBBR	SBBH				
C05	Liver	SBBH	SBVT	SBRJ	SBCT	SBRF	SBPA				
C06	Kidneys	SBBH	SBRJ	SBVT	SBPA						
C07	Liver	SBCG	SBRF	SBBR	SBRJ	SBVT	SBBH	SBPA			
C08	Heart	SBCG	SBSP	SBBR							
C09	Kidneys	SBCG	SBSP	SBEG	SBBE	SBBR	SBGO	SBRF			
C10	Liver	SBSV	SBRF	SBFZ	SBRJ	SBCT	SBVT	SBBR	SBBH		
C11	Kidney	SBRB	SBRF	SBBR	SBBE	SBGO					
C12	Kidney	SBRB	SBRF	SBGR	SBBR	SBBE	SBGO				
C13	Liver	SBRB	SBBR	SBRJ	SBVT	SBRF	SBFZ				
C14	Liver	SBNT	SBSV	SBRF	SBFZ	SBRJ	SBBR	SBCT	SBVT		
C15	Kidney	SBFL	SBSP	SBPA	SBCT	SBRJ	SBRF	SBBH	SBBE	SBVT	
C16	Kidney	SBFL	SBPA	SBCT	SBRJ	SBRF	SBBH	SBBE	SBVT		
C17	Kidney	SBFL	SBRF	SBSP	SBPA						
C18	Liver	SBNT	SBRF	SBFZ	SBRJ	SBBR	SBVT	SBCT			
C19	Liver	SBNT	SBGR	SBRF	SBFZ	SBCT	SBRJ	SBBR	SBVT		
C20	Liver	SBSV	SBFZ	SBRF	SBRJ	SBBR					
C21	Kidney	SBFZ	SBSV	SBMO	SBPA	SBRJ	SBBE	SBTE	SBJP	SBRF	SBSL
C22	Kidneys	SBFZ	SBSV	SBMO	SBJP	SBRF	SBNT	SBTE			
C23	Heart	SBGO	SBBR								
C24	Lung	SBGO	SBBR								
C25	Liver	SBGO	SBBR	SBRJ	SBVT	SBPE	SBCE	SBPR			

Table 2
Composition of times and values.

Organ	CIT_{max}	D_{eg}	D_{hs}	D_{th}	D_{max}
Heart	04:00	00:30	00:30	00:30	02:30
Lung	06:00	00:30	00:30	00:30	04:30
Liver	12:00	00:40	00:30	00:30	10:20
Pancreas	20:00	01:00	00:30	00:30	18:00
Kidney	36:00	01:20	00:30	00:30	33:40

origin airport to the destination airport (D_{st}) is analyzed, the reduction increased to 44,17% on average.

For better visualization of the results presented in Table 3, Fig. 2 illustrates the paths for cases where the routes differ between the real path (RP) and the shortest path (SPA), specifically C01, C06, C07, C14, C16, C20, and C22. The RP, constructed by the CNT, is represented by a dashed line, while the SPA, provided by the labeling algorithm, is shown as a solid line.

Although these results are promising, they were already achieved in Carrara et al. (2018). The main contribution of this work resides on the fact that the algorithm can compute the shortest path between all airports in a fraction of the time required to solve a single instance of the mathematical model with the aid of a commercial solver. The algorithm can almost instantly return different solutions in respect to the penalty P and ensure optimality. In addition, it is free and does not incur additional costs, such as the cost of a solver license acquisition by the CNT.

The Table 4 shows the paths found by the dynamic programming algorithm for all instances and the instant of time when the organ was available for transport at the origin airport ($H_{available}$). It also shows the arrival time in the destination airport (T_i), the difference between these times, which correspond to the transport time (D_{st}), and the execution time of the dynamic programming algorithm. It can be observed that the set of instances, on which the algorithm is tested, corresponds to all Cases presented in Table 1. For each Case, instances were generated from the origin to destination airports in increasing order of priority, up to the real destination chosen by CNT was reached.

The results from Table 4 can be compared with the results in Carrara et al. (2018). The authors report times of execution varying from 5 s to 10 min, and they also report optimization GAPs for some instances.

Then, for the sake of comparison, the performance of the algorithm will be compared with the upper bound.

The results for most instances are the same, although it is not possible to know if CPLEX proved optimality within the 10 min execution time. The dynamic programming algorithm proposed, however, has found the optimal solution always equal or better than the obtained by the solver, almost instantly. The Table 5 shows cases in which the solution found by the algorithm differs from those obtained with the aid of CPLEX. Since these latter solutions are not optimal, it can be deduce that the solver ran for 10 min and provided a suboptimal solution. However, it is important to remind that this time value (600 s) must be multiplied by the computers conversion factor, equal to 0.3. Thus resulting in execution times of 180 s.

5.3. The effect of the penalty

As stated before, for Tables 3–5, all solutions presented were calculated with the penalty parameter P equal to 30 min. For instance, one can observe that in Table 5, the solutions provided by the algorithm for instances C01 SBRJ and C22 SBMO reach the final airport destination later when compared to the solutions provided by CPLEX. However, the solutions found by CPLEX prescribe more flights, and therefore represent suboptimal solutions.

It is important to remember that the addition of the penalty intended to give solutions a better balance between arrival time and the number of flights. To better visualize this effect when choosing the optimal solution, the Table 6 shows the solutions with the penalty P equals to zero and 30 min, in which they differ. To facilitate the understanding of choosing one solution over the other, column f_{obj} shows the value of the solution. This is calculated by summing the arrival time in the destination airport (T_i) with the number of flights multiplied by the value of the penalty P , as in Objective Function (1).

As the number of connections flights required to traverse the path from the origin to the destination airport increases, so does the chance of an unwanted event such as a delay or even a flight cancellation. Moreover, flight connections require more handling of the organ, which also must be avoided. Table 6 shows how the penalty can help in providing better solutions, by avoiding flight connections when possible.

Table 3
Comparison between the paths built by CNT and the ones provided by the algorithm for the same destination.
Source: Adapted from Carrara et al. (2018).

Real path				Shortest path (algorithm)				
Case	Path	D_{st}	CIT	Path	D_{st}	CIT	Diff.	CIT Red. (%)
C01	SBSV-SBBH-SBVT	08:11	10:31	SBSV-SBGR-SBVT	05:18	07:38	02:53	27,42
C02	SBRJ-SBPA	17:42	20:02	SBRJ-SBPA	02:38	04:58	15:03	75,12
C03	SBFZ-SBRF	09:51	12:11	SBFZ-SBRF	03:55	06:15	05:55	48,56
C04	SBSV-SBGR-SBPA	19:50	22:10	SBSV-SBGR-SBPA	04:35	06:55	15:14	68,72
C05	SBBH-SBVT	07:03	08:43	SBBH-SBVT	05:29	07:09	01:34	17,97
C06	SBBH-SBSP-SBPA	06:51	09:11	SBBH-SBPA	05:35	07:55	01:15	13,61
C07	SBCG-SBBR	FAB	Aircraft	SBCG-SBKP-SBBR	05:25	07:05	N/A	N/A
C08	SBCG-SBSP	Charter aircraft		Infeasible	N/A	N/A	N/A	N/A
C09	SBCG-SBSP	Charter aircraft		SBCG-SBSP	10:35	12:55	N/A	N/A
C10	SBSV-SBRF	06:21	08:01	SBSV-SBRF	04:52	06:32	01:28	18,30
C11	SBRB-SBBR-SBRF	09:18	11:38	SBRB-SBBR-SBRF	09:18	11:38	00:00	0,00
C12	SBRB-SBBR-SBGR	08:30	10:50	SBRB-SBBR-SBGR	08:13	10:33	00:16	2,46
C13	SBRB-SBBR	06:45	08:25	SBRB-SBBR	06:41	08:21	00:03	0,59
C14	SBNT-SBFZ	06:37	08:17	SBNT-SBRF-SBFZ	04:39	06:19	01:58	23,74
C15	SBFL-SBSP	15:44	18:04	SBFL-SBSP	01:35	03:55	14:08	78,23
C16	SBFL-SBPA-SBCT	16:47	19:07	SBFL-SBSP-SBCT	03:13	05:33	13:33	70,88
C17	SBFL-SBSP	17:10	19:30	SBFL-SBSP	03:11	05:31	13:58	71,62
C18	SBNT-SBFZ	03:54	05:34	SBNT-SBFZ	03:43	05:23	00:10	2,99
C19	SBNT-SBGR	06:32	08:12	SBNT-SBGR	04:22	06:02	02:10	26,42
C20	SBSV-SBNT-SBFZ	04:07	05:47	SBSV-SBFZ	02:15	03:55	01:51	31,99
C21	SBFZ-SBGR-SBPA	20:05	22:25	SBFZ-SBGR-SBPA	06:20	08:40	13:44	61,26
C22	SBFZ-SBBR-SBTE	21:33	23:53	SBFZ-SBTE	04:23	06:43	17:09	71,81
C23	SBGO-SBBR	FAB	Aircraft	SBGO-SBBR	02:24	03:54	N/A	N/A
C24	SBGO-SBBR	FAB	Aircraft	SBGO-SBBR	02:24	03:54	N/A	N/A
C25	SBGO-SBBR	FAB	Aircraft	SBGO-SBBR	02:14	03:54	N/A	N/A

N/A - Not Available. It was not possible to calculate the Difference column because there is no data available on Charter or Military flights.

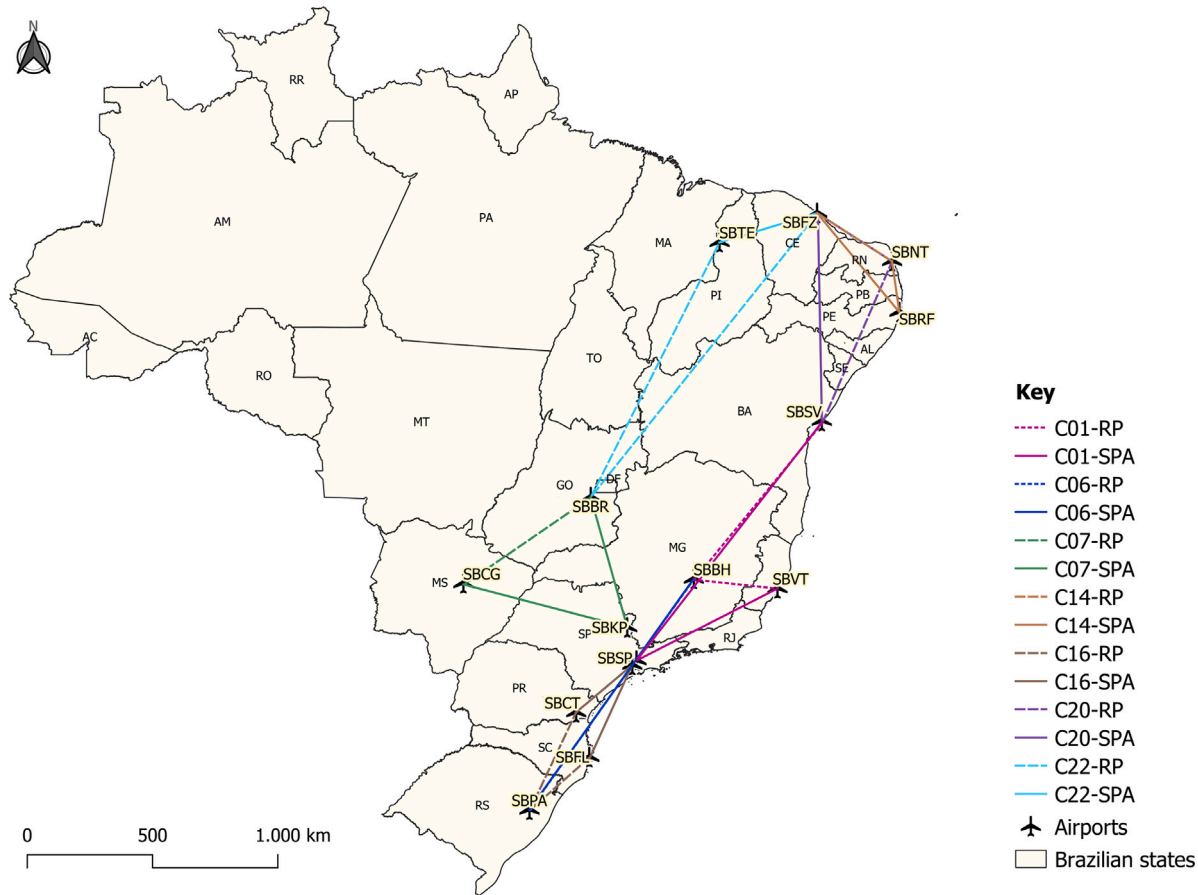


Fig. 2. Path differences between CNT and labeling algorithm.

Table 4
Results and execution time of the labeling algorithm.

Instance	Path	$H_{available}$	T_i	D_{st}	Execution time (s)
C01 SBRF	SBSV-SBRF	02:42	07:27	04:45	0.002050
C01 SBFZ	SBSV-SBRF-SBFZ	02:42	09:24	06:42	0.002061
C01 SBRJ	SBSV-SBRJ	02:42	07:23	04:41	0.001948
C01 SBVT	SBSV-SBGR-SBVT	02:42	08:00	05:18	0.001973
C02 SBVT	SBRJ-SBVT	06:49	08:23	01:33	0.002354
C02 SBBH	SBRJ-SBBH	06:49	08:18	01:28	0.001976
C02 SBRF	SBRJ-SBRF	06:49	10:21	03:31	0.001912
C02 SBCT	SBRJ-SBCT	06:49	08:18	01:28	0.002014
C02 SBPA	SBRJ-SBPA	06:49	09:28	02:38	0.002058
C03 SBRF	SBFZ-SBRF	09:34	13:29	03:54	0.001884
C04 SBRF	SBSV-SBRF	13:34	15:48	02:13	0.002108
C04 SBFZ	SBSV-SBFZ	13:34	15:10	01:35	0.001885
C04 SBPA	SBSV-SBGR-SBPA	13:34	18:10	04:35	0.001503
C05 SBVT	SBBH-SBVT	02:52	08:21	05:28	0.001015
C06 SBRJ	SBBH-SBRJ	03:31	07:09	03:37	0.001964
C06 SBVT	SBBH-SBVT	03:31	08:21	04:49	0.002006
C06 SBPA	SBBH-SBPA	03:31	09:07	05:35	0.002192
C07 SBRF	SBCG-SBGR-SBRF	18:45	02:03 ^a	07:18	0.001752
C07 SBBR	SBCG-SBKP-SBBR	18:45	00:10 ^a	05:25	0.001542
C08 SBGR	Infeasible	18:34	N/A	N/A	0.000113
C08 SBKP	SBCG-SBKP	18:34	20:33	01:58	0.000115
C08 SBRP	Infeasible	18:34	N/A	N/A	0.000123
C08 SBSP	Infeasible	18:34	N/A	N/A	0.000117
C09 SBGR	SBCG-SBGR	19:25	21:22	01:57	0.001951
C09 SBKP	SBCG-SBGR-SBCY-SBKP	19:25	04:42 ^a	09:17	0.002147
C09 SBRP	SBCG-SBGR-SBRP	19:25	23:16	03:51	0.002223
C09 SBSP	SBCG-SBSP	19:25	06:00 ^a	10:35	0.002646
C10 SBRF	SBSV-SBRF	02:34	07:27	04:53	0.001046
C11 SBRF	SBRB-SBBR-SBRF	01:30	10:48	09:18	0.001809
C12 SBRF	SBRB-SBBR-SBRF	01:30	10:48	09:18	0.001951
C12 SBSP	SBRB-SBBR-SBSP	01:30	09:58	08:28	0.001922
C12 SBKP	SBRB-SBBR-SBKP	01:30	09:52	08:22	0.001874
C12 SBRP	SBRB-SBBR-SBGR-SBRP	01:30	11:57	10:27	0.002136
C12 SBGR	SBRB-SBBR-SBGR	01:30	09:43	08:13	0.001978
C13 SBBR	SBRB-SBBR	00:49	07:31	06:41	0.000399
C14 SBSV	SBNT-SBRF-SBSV	04:45	08:52	04:07	0.001468
C14 SBRF	SBNT-SBRF	04:45	07:22	02:37	0.001118
C14 SBFZ	SBNT-SBRF-SBFZ	04:45	09:24	04:39	0.001192
C15 SBGR	SBFL-SBGR	16:19	18:10	01:50	0.001870
C15 SBKP	SBFL-SBKP	16:19	21:12	04:52	0.001769
C15 SBRP	SBFL-SBSP-SBRP	16:19	20:16	03:57	0.001831
C15 SBSP	SBFL-SBSP	16:19	17:56	01:36	0.001983
C16 SBPA	SBFL-SBPA	16:19	18:33	02:13	0.002748
C16 SBCT	SBFL-SBSP-SBCT	16:19	19:33	03:14	0.002059
C17 SBRF	SBFL-SBGR-SBRF	14:04	21:37	07:33	0.002546
C17 SBGR	SBFL-SBGR	14:04	16:42	02:37	0.002113
C17 SBKP	SBFL-SBKP	14:04	21:12	07:07	0.002219
C17 SBRP	SBFL-SBSP-SBRP	14:04	20:16	06:12	0.002969
C17 SBSP	SBFL-SBSP	14:04	17:16	03:11	0.001918
C18 SBRF	SBNT-SBRF	11:48	16:16	04:28	0.000760
C18 SBFZ	SBNT-SBFZ	11:48	15:31	03:43	0.000839
C19 SBSP	SBNT-SBRJ-SBSP	00:52	07:31	06:38	0.000744
C19 SBKP	SBNT-SBKP	00:52	05:33	04:40	0.000685
C19 SBRP	SBNT-SBGR-SBRP	00:52	08:52	07:59	0.000716
C19 SBGR	SBNT-SBGR	00:52	05:15	04:22	0.000700
C20 SBFZ	SBSV-SBFZ	12:55	15:10	02:14	0.000914
C21 SBSV	SBFZ-SBRF-SBSV	16:49	20:49	04:00	0.002304
C21 SBMO	SBFZ-SBRF-SBMO	16:49	20:42	03:52	0.002255
C21 SBPA	SBFZ-SBGR-SBPA	16:49	23:10	06:20	0.002330
C22 SBSV	SBFZ-SBNT-SBSV	01:49	06:04	04:15	0.002222
C22 SBMO	SBFZ-SBGR-SBMO	01:49	08:57	07:08	0.001899
C22 SBJP	SBFZ-SBRJ-SBJP	01:49	11:00	09:10	0.001959
C22 SBRF	SBFZ-SBNT-SBRF	01:49	07:22	05:33	0.001871
C22 SBNT	SBFZ-SBNT	01:49	03:10	01:20	0.001758
C22 SBTE	SBFZ-SBTE	01:49	06:13	04:24	0.002068
C23 SBBR	SBGO-SBBR	04:36	07:00	02:24	0.000121
C24 SBBR	SBGO-SBBR	04:36	07:00	02:24	0.000339
C25 SBBR	SBGO-SBBR	04:46	07:00	02:13	0.001378

^a The organ arrives the day after departure.

Table 5
Paths and time comparison between CPLEX and algorithm solutions, respectively.

Instance	MILP formulation with ILOG CPLEX 12.5			Labeling algorithm		
	Path	T_i	Time (s)	Path	T_i	Time (s)
C01 SBRJ	SBSV-SBKP-SBRJ	07:22	180 ^b	SBSV-SBRJ	07:23	0.001948
C07 SBRF	No viable solution found	N/A	180 ^b	SBCG-SBGR-SBRF	02:03 ^a	0.001752
C09 SBKP	SBCG-SBKP	12:10 ^a	180 ^b	SBCG-SBGR-SBCY-SBKP	04:42 ^a	0.002147
C17 SBKP	SBFL-SBPA-SBKP	21:16	180 ^b	SBFL-SBKP	21:12	0.002219
C22 SBMO	SBFZ-SBNT-SBRF-SBMO	08:40	180 ^b	SBFZ-SBGR-SBMO	08:57	0.001899
C22 SBJP	SBFZ-SBNT-SBSV-SBMO-SBJP	11:00	180 ^b	SBFZ-SBRJ-SBJP	11:00	0.001959

^a The organ arrives the day after departure.

^b The computational times of Carrara et al. (2018) were multiplied by 0.3 in order to fairly compare the results, as mentioned previously.

Table 6
The effect of the penalty in solutions.

$P = 0$					$P = 30$ min			
Instance	Path	T_i	Flights	f_{obj}	Path	T_i	Flights	f_{obj}
C01 SBRJ	SBSV-SBKP-SBRJ	07:22	2	07:22	SBSV-SBRJ	07:23	1	07:53
C07 SBRF	SBCG-SBKP-SBRJ-SBRF	01:53 ^a	3	01:53 ^a	SBCG-SBGR-SBRF	02:03 ^a	2	03:03 ^a
C09 SBSP	SBCG-SBGR-SBCY-SBSP	05:57 ^a	3	05:57 ^a	SBCG-SBSP	06:00 ^a	1	06:30 ^a
C17 SBKP	SBFL-SBSP-SBRJ-SBKP	20:23	3	20:23	SBFL-SBKP	21:12	1	21:42
C22 SBMO	SBFZ-SBNT-SBRF-SBMO	08:40	3	08:40	SBFZ-SBGR-SBMO	08:57	2	09:57

^a The organ arrives the day after departure.

The addition of the penalty in the objective function provides an optimal path that arrives just one minute later for the instance C01 SBRJ. However, it uses one less flight in comparison with the optimal solution for $P = 0$. In that sense, the effect of the penalty in instance C09 SBSP is even more dramatic. Instead of providing the earliest arrival time in a chain of 3 flights for $P = 0$, the labeling algorithm with $P = 30$ min provides a solution that uses one direct flight from the origin to the destination airport with the cost of arriving just 3 min later.

As travel time is very important due to the organs' perishability, the faster the transfer, the greater the chances of the transplant going well. Therefore, if we weigh up the penalty too heavily (e.g., making $P = 60$ or $P = 100$), we may miss solutions where it would be worth taking an extra flight to get to the destination sooner. In addition, there is an operational motivation for considering $P = 30$ as well, since this is the average time expected to be spent by changing aircraft within the same airport.

At last, the algorithm developed in this work, as implemented, is not as sensitive to small or large changes in the input parameters. Whenever there are changes (big or not) in the input parameters, the algorithm can be re-run without major difficulties.

6. Conclusions

This paper presented a labeling algorithm to solve the transport of organs for transplantation problem. The algorithm minimizes the arrival time and the total number of flights used. This approach differs from the previous attempts on the literature to solve this problem, since it does not depend on a commercial solver and can provide the optimal path (if one exists) with low execution times.

As reported in Carrara et al. (2018), the commercial solver CPLEX could not solve some instances to optimality within 10 min of execution. It is also reported that for some instances CPLEX ran more than two hours without finding the optimal solution. However, our implementation has found the optimal solution for all instances.

The algorithm's solutions, when compared with the routes taken in reality, shows, on average, a potential to reduce the CIT by 37,46% and the transport time by 44,17%. The results also show that the automation of the task performed by CNT technicians could lead in a more fair choice of the receiver in respect to the priority list.

The labeling algorithm can solve the problem efficiently and could have a great impact on the quality of the job performed by CNT

technicians and in the post intervention life of organ receivers. Naturally, a future step is the implementation of this algorithm in the CNT system, which requires the development of an user-friendly graphical user interface, in which technicians can easily enter data such as organ, origin and a ranked list of destinations. As mentioned before, the algorithm can be adapted to check if it is possible to transport an organ for a potential recipient and, in the absence of a feasible chain of flights, check the next candidate, until a recipient is found.

In that sense, it is important to reaffirm that the algorithm could return paths between multiple origin–destination pairs without having to be recalculated. Since the CNT system has access to all flights being operated between Brazilian airports, the number of nodes and arcs is expected to have a sharp increase in comparison with the instances proposed in Carrara et al. (2018), consequently requiring higher execution times. In order to investigate the performance of the algorithm in larger air transport networks, a next and necessary step is measuring the performance of the algorithm in practice.

In addition, as seen in the comparison between the paths built by CNT and the ones provided by the algorithm for the same destination, in one case the recipient could not be reached with commercial flights but the organ was transported in feasible time by a charter aircraft. Before offering an organ to a next recipient, it is necessary to contact the current candidate and ask if he is able to cover the transport costs in order to be transplanted. This is useful in restricted operating scenarios but may require further adjustments of the algorithm before its release in the CNT system.

Finally, the proposed algorithm represents a valuable opportunity to enhance the CNT system by automating complex and time-sensitive tasks in organ transport planning, which are currently performed manually. Its integration could improve decision-making processes, ensure more efficient use of resources, and increase the chances of success. However, as the CNT is a public institution in Brazil, the implementation of the algorithm may require more time to accommodate institutional requirements and regulatory procedures. Furthermore, the implementation of the algorithm will support further research on organ transport and its effects on transplantation rates and outcomes in the medium and long term.

CRedit authorship contribution statement

Isaac Balster: Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Joyce Azevedo Caetano:** Writing

– review & editing. **Glaydston Mattos Ribeiro:** Writing – review & editing, Supervision. **Laura Bahiense:** Writing – review & editing, Supervision.

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Data availability

The authors do not have permission to share data.

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